



7806 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging compact object in SN 1987A with JWST

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
MIRI				
	2	MRS - BKG	MIRI Medium Resolution Spectroscopy	(2) SN1987A-BKG-SE
	1	MRS	MIRI Medium Resolution Spectroscopy	(1) SN-1987A
NIRSpec				
	3	NIRSpec IFU - G140H	NIRSpec IFU Spectroscopy	(3) SN-1987A-nirspec
	4	NIRSpec IFU - G235H/ G395H	NIRSpec IFU Spectroscopy	(3) SN-1987A-nirspec
	5	NIRSpec IFU - G140M /G395M	NIRSpec IFU Spectroscopy	(3) SN-1987A-nirspec

ABSTRACT

SN 1987A in the Large Magellanic Cloud is the closest observed supernova (SN) in centuries and provides a truly unique opportunity to probe various aspects of SN physics and evolution. SN 1987A has been continuously scrutinized as it changes significantly year after year providing new discoveries and insights into the evolution of SNe in general. Previous JWST observations revealed the onset, and early evolution of the interaction between the rapidly expanding, metal-rich inner ejecta and the circumstellar equatorial ring (ER), a new phase in the evolution of the remnant. The observations also revealed the first clear evidence of the compact object created in the explosion. The goals of this proposal are to determine the metal-rich ejecta structure and composition in the interaction regions, and monitor the emerging compact object and evolution of dust in the outer ejecta. JWST NIRSpec and MIRI give access to numerous important diagnostic lines for the ejecta-ER interaction, the ejecta morphology, and is the only telescope that has unambiguously detected emission associated with the compact object. The science objectives rely on analyzing time variations compared to the previous JWST observations. By Cycle 4, the ejecta and the interaction regions will have evolved substantially in the two years since the last JWST observations. The proposed observations also hold great legacy value for the community. SN 1987A has been observed regularly by all major telescopes since the very beginning, which has created a unique observational record of the evolution of a SN into a SN remnant across the electromagnetic spectrum.

OBSERVING DESCRIPTION

NIRSpec/IFU and MIRI/MRS will be used to observe SN1987A at a third epoch with continuous spectral coverage from 1 to 28 microns.

NIRSpec/IFU observations use the G140H/F100LP, G235H/F170LP, G395H/F290LP, and G140H/F070LP with a 4-point dither on the SMALL CYCLING pattern. The high resolution gratings are required to disentangle all the emission components, including the compact object. Observations with G140M/F100LP G395M/F290W disperser/filter combinations with a 4-point dither on the SMALL CYCLING pattern are taken to cover the wavelength gaps of G140H and G395 which include the [Fe I] 1.44 micron line and the continuum near the [Ca V] 4.16 micron line, crucial to our science goals.

The telescope PA is constrained based on the Cycle 2 observation to 65.5 (+/-10) degrees so that the leakcal from Cycle 2 observations of SN 1987A can be used to assess contamination by nebosity in this region. Point source contamination is easily removed either with the updated outlier rejection in the pipeline or manual masking. The readout pattern used is NRSIRS2RAPID for all NIRCAM exposures.

MIRI/MRS observations use the SHORT, MEDIUM and LONG dispersers to obtaining the full 4.9-28 micron coverage with a 4-point dither on the EXTENDED SOURCE pattern. The SLOWR1 readout pattern is used with 1 integrations of 49 groups at each dither position across all dispersers on

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each detectors. A dedicated background to the south of SN1987A is taken to subtract from our science exposures and correct for detector effects such as warm pixels and residual dark current. A 4-point dither on the EXTENDED SOURCE pattern is used with a reduced number of integrations compared to the science observations, namely 1 integrations of 20 groups at each dither, which is sufficient to characterise and remove the background. SLOWR1 is necessary to avoid data excess exceeding the middle threshold.

The MIRI/MRS on-source and background observations are to be carried out in a non-interruptible sequence to ensure the background is representative of the source field.

The MIRI/MRS and NIRSpec/IFU observations are ‘Grouped within 14 days’ to minimise variability.

The blind pointing accuracy of JWST is sufficient for this programme so no TA is requested. SN 1987A is in the JWST continuous viewing zone.

Proposal 7806 - Targets - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging compact o...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN-1987A	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000		
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae] Extended=YES				
	(2)	SN1987A-BKG-SE	RA: 05 35 28.5915 (83.8691312d) Dec: -69 16 20.99 (-69.27250d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(3)	SN-1987A-nirspec	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000		
	Comments: This entry is a duplicate of Source 1 SN-1987A only without a backgroud specification to allow NIRSpec observations with no dedicated background. Category=Star Description=[Supernovae] Extended=YES				

Proposal 7806 - Observation 2 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Observation	Proposal 7806, Observation 2: MRS - BKG													Tue Mar 11 21:06:59 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MRS (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(2)	SN1987A-BKG-SE	RA: 05 35 28.5915 (83.8691312d) Dec: -69 16 20.99 (-69.27250d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	F560W	All MRS				NO			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
	2	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		SLOWR1	20	1	1	Dither 1	4	4	1911.194		
	1	LONG(C)	MRSSHORT		SLOWR1	20	1	1	Dither 1	4	4	1911.194		
	2	MEDIUM(B)	MRSLONG		SLOWR1	20	1	1	Dither 1	4	4	1911.194		
	2	MEDIUM(B)	MRSSHORT		SLOWR1	20	1	1	Dither 1	4	4	1911.194		
	3	SHORT(A)	MRSLONG		SLOWR1	20	1	1	Dither 1	4	4	1911.194		
	3	SHORT(A)	MRSSHORT		SLOWR1	20	1	1	Dither 1	4	4	1911.194		

Proposal 7806 - Observation 2 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Special Requirements	Sequence Observations 1, 2, Non-interruptible Group Observations 1, 2, 3, 4, 5 within 14 Days
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Proposal 7806 - Observation 1 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Observation	Proposal 7806, Observation 1: MRS												Tue Mar 11 21:06:59 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MRS - BKG (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	SN-1987A	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Star												
	Description=[Supernovae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F560W	All MRS			NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	LONG(C)	MRSLONG		SLOWR1	49	1	1	Dither 1	4	4	4682.424	
	1	LONG(C)	MRSSHORT		SLOWR1	49	1	1	Dither 1	4	4	4682.424	
	2	MEDIUM(B)	MRSLONG		SLOWR1	49	1	1	Dither 1	4	4	4682.424	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	49	1	1	Dither 1	4	4	4682.424	
	3	SHORT(A)	MRSLONG		SLOWR1	49	1	1	Dither 1	4	4	4682.424	
	3	SHORT(A)	MRSSHORT		SLOWR1	49	1	1	Dither 1	4	4	4682.424	

Proposal 7806 - Observation 1 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Special Requirements	Sequence Observations 1, 2, Non-interruptible Group Observations 1, 2, 3, 4, 5 within 14 Days
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Proposal 7806 - Observation 3 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Observation	Proposal 7806, Observation 3: NIRSpec IFU - G140H											Tue Mar 11 21:06:59 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	SN-1987A-nirspec	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000									
Comments: This entry is a duplicate of Source 1 SN-1987A only without a background specification to allow NIRSpec observations with no dedicated background. Category=Star Description=[Supernovae] Extended=YES												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		4				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	NRSIRS2RAPID	105	1	false	true	NONE	4	4	6185.689	
	2	G140H/F070LP	NRSIRS2RAPID	105	1	false	true	NONE	4	4	6185.689	
Special Requirements	Aperture PA Range 55.5 to 75.5 Degrees (V3 276.52835083 to 296.52835083)											
	Group Observations 1, 2, 3, 4, 5 within 14 Days											

Proposal 7806 - Observation 4 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Observation	Proposal 7806, Observation 4: NIRSpec IFU - G235H/G395H											Tue Mar 11 21:06:59 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over lower threshold											
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	SN-1987A-nirspec	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000									
	Comments: This entry is a duplicate of Source 1 SN-1987A only without a background specification to allow NIRSpec observations with no dedicated background. Category=Star Description=[Supernovae] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size	Starting Point		Number of Points		Points			
	1	CYCLING		SMALL	1		4					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	77	1	false	true	NONE	4	4	4551.734	
	2	G395H/F290LP	NRSIRS2RAPID	63	2	false	true	NONE	4	8	7469.512	
Special Requirements	Aperture PA Range 55.5 to 75.5 Degrees (V3 276.52835083 to 296.52835083)											
	Group Observations 1, 2, 3, 4, 5 within 14 Days											

Proposal 7806 - Observation 5 - The infancy of a supernova remnant: Probing ejecta dynamics, dust evolution, and the emerging com...

Observation	Proposal 7806, Observation 5: NIRSpec IFU - G140M/G395M											Tue Mar 11 21:06:59 GMT 2025	
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	SN-1987A-nirspec	RA: 05 35 27.9680 (83.8665333d) Dec: -69 16 11.09 (-69.26975d) Equinox: J2000										
	Comments: This entry is a duplicate of Source 1 SN-1987A only without a background specification to allow NIRSpec observations with no dedicated background. Category=Star Description=[Supernovae] Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		4					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G140M/F100LP	NRSIRS2RAPID	29	1	false	true	NONE	4	4	1750.667		
	2	G395M/F290LP	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467		
Special Requirements	Aperture PA Range 55.5 to 75.5 Degrees (V3 276.52835083 to 296.52835083)												
	Group Observations 1, 2, 3, 4, 5 within 14 Days												